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FFGFT — Falsifiability

Physical Theory vs. Metaphysical Speculation

Companion document to Dok. 248

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This document clarifies the philosophy-of-science position of FFGFT. FFGFT is a physical theory with empirical content and concrete falsification conditions (L_0, D_6, singularities, α). The postulate of "pure, unrestrained information" (Myser 2026) is a metaphysical framing without derivation power — not falsifiable, not connectable to known physics.

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1. Falsifiability as a Scientific Criterion

A scientific theory distinguishes itself from metaphysical speculation through its **vulnerability to refutation** [Popper 1934].

> **Popper's Criterion:** A theory is scientific if it is **principally falsifiable** — if there exist observations that would contradict the theory and thereby refute it. A theory compatible with every possible observation says nothing about the world.

2. FFGFT Is Falsifiable

2.1. Minimal Length Scale L_0

$$L_0 = \xi \cdot \square\_P \approx 2.2 \times 10^{-39} \text{ m}$$

If experiments consistently showed structures *below* this scale, FFGFT would be falsified.

2.2. Fractal Dimension D_6

$$D_6 = 2.999867$$

Deviation from exactly 3 by 1.33×10^{-4}. Measurable via **gamma-ray burst time-of-flight dispersion** and **gravitational wave phase structure**. An incompatible measurement would falsify FFGFT.

2.3. No Singularity in Black Holes

L_0 sets an absolute density bound. Distinguishable from GR through **gravitational wave echoes** and a modified **Hawking radiation spectrum**.

2.4. Coupling Constants from ξ

$\alpha^{-1} = 137.036$ (deviation < 0.001 %) · a_e (K_frak^{3e2} reduces deviation 2.03 % →0.014 %) · Koide formula (geometric derivation).

3. The Postulate of “Pure Information” Is Not Falsifiable

3.1. No Derivable Quantities

Without structure no derivation, without derivation no prediction.

3.2. Compatible With Every Observation

Since nothing follows from the postulate, it cannot be refuted by anything.

> **Not falsifiable = not physical.** The postulate of “pure information” is a **metaphysical framing** — legitimate, but of a fundamentally different kind from FFGFT.

3.3. No Occlusion Mechanism Derivable

Atticus's O-bit adhesion rule is itself already structure imported from outside. The primitive does not carry the burden of occlusion.

4. The Philosophy-of-Science Distinction

Criterion	FFGFT	"Pure information" (Myser)
Primitive	$T^4 + \xi$ (geometric, posited)	"pure unrestrained information" (posited)
Operational	Yes — derivations, field equations, measurement	No — no derivation possible
Falsifiable	Yes — L_0, D_6, singularities, α	No — no observation excluded
Predictions	Concrete, numerical	None
Connectable	To Standard Model, GR, QFT	Not directly
Type of claim	Physical theory	Metaphysical framing

The decisive distinction: FFGFT is a physical theory with empirical content. The postulate of "pure information" is a metaphysical speculation without derivation power. The difference lies not only in the axiom choice, but in the *consequences* of that choice.

5. Falsifiability and Epistemic Openness

This distinction is not an attack on metaphysical framings. Metaphysical questions are legitimate and important — simply not decidable by physical experiments. FFGFT leaves "why T^4?" explicitly open (cf. Dok. 248).

FFGFT is falsifiable but epistemically modest. It makes no claims about what lies before T^4. It makes precise claims about what follows from $T^4 + \xi$ — and these claims can be refuted by observation.

Summary

Claim	Position
FFGFT makes no predictive claims	False — L_0, D_6, singularities, α are concrete predictions
FFGFT is not falsifiable	False — every prediction is in principle refutable
"Pure information" is a physical theory	No — metaphysical framing without derivation power

Claim	Position
“Pure information” is falsifiable	No — no observation is excluded
Both approaches are scientifically equivalent	No — only FFGFT is a physical theory in Popper’s sense
Metaphysical framings are worthless	No — legitimate and important, but of a fundamentally different kind
FFGFT answers “why T^4?”	No — remains explicitly open; structural honesty (cf. Dok. 248)

References

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